

# Marketing Analytics Datathon

## Excel + Python Playbook

A practical guide for rebuilding and explaining the project

### Big idea

- ? Use Python as the source of truth for repeatable cleaning, joins, attribution, machine learning, dashboard generation, validation, and the PowerPoint deck.
- ? Use Excel for inspection, PivotTables, sanity checks, business explanation, and optional workbook dashboards.
- ? Do not force the full project into Excel. Some parts are possible but fragile, especially attribution and machine learning.

### Final outputs to know

- ? Dashboard: `public/index.html`
- ? Generated dashboard: `outputs/dashboard/Marketing_Analytics_Dashboard.html`
- ? Deck: `outputs/presentation/Marketing_Analytics_Executive_Deck_v4.pptx`
- ? Excel reports: `outputs/analysis/*.xlsx`
- ? Validator: `analytics_case_study/06_validate_metrics.py`

# 1. What Python Does

## Run these scripts in order

- ? 01\_data\_cleaning.py - reads raw Excel files, cleans columns, normalizes domains, deduplicates opportunity/account logs, writes cleaned parquet files.
- ? 02\_data\_integration.py - joins account, opportunity, campaign, email, web, ICP, and segment data into analysis-ready tables.
- ? 03\_analysis.py - creates Excel reports for channel ROI, segments, creative performance, email performance, and budget scenarios.
- ? 03b\_attribution.py - builds touchpoints and attribution models: sourced, influenced, first-touch, last-touch, linear, and time-decay.
- ? 03c\_advanced\_analytics.py - builds win probability model, account coverage, deal velocity, journey sequences, targeting matrix, and cohorts.
- ? 04\_html\_dashboard.py - generates the interactive HTML dashboard and copies it to public/index.html.
- ? 05\_presentation.py - generates the executive PowerPoint deck.
- ? 06\_validate\_metrics.py - checks that required outputs exist and dashboard files are in sync.

## Why Python is needed

- ? It makes the workflow repeatable.
- ? It handles large joins and row-level attribution without manual spreadsheet errors.
- ? It can train the Random Forest model, which native Excel cannot reproduce.
- ? It validates outputs before submission.

## 2. What Excel Is Best For

### Use Excel to review generated reports

- ? Open outputs/analysis/channel\_roi.xlsx for channel pipeline, win rate, spend, and ROI.
- ? Open outputs/analysis/segment\_conversion.xlsx for segment, industry, and seniority views.
- ? Open outputs/analysis/creative\_performance.xlsx for ad CTR, CPC, CPM, and creative patterns.
- ? Open outputs/analysis/email\_campaign\_performance.xlsx for email open/click/register performance.
- ? Open outputs/analysis/attribution\_models.xlsx for attribution model comparison.
- ? Open outputs/analysis/advanced\_analytics.xlsx for coverage, velocity, journey, cohort, targeting, and ML outputs.

### Use Excel to build optional pivots

- ? Pivot by channel\_category to explain pipeline and win rate.
- ? Pivot by segment\_\_c and industry to explain target markets.
- ? Pivot by coverage\_tier to show unreached account opportunity.
- ? Pivot by \_seniority to explain email/ICP audience quality.
- ? Pivot by quarter to show pipeline growth and win-rate decline.

# 3. Parts You Should NOT Build Fully In Excel

## Multi-touch attribution

- ? Excel can approximate it, but the project links tens of thousands of touchpoints to opportunities within a 365-day lookback window.
- ? First-touch and last-touch require sorting each opportunity path.
- ? Linear attribution requires counting unique channels per opportunity.
- ? Time-decay attribution requires per-touchpoint weights and normalized credit per opportunity.
- ? Use Python output, then review attribution\_models.xlsx in Excel.

## Journey sequences

- ? Ordered paths like email\_mqa -> 6sense\_display are hard to maintain in formulas.
- ? Python creates clean sequence outputs. Excel should only review the final table.

## Win probability model

- ? Native Excel cannot train the same Random Forest model used here.
- ? Excel can create a simple scoring model, but that is only an approximation.
- ? Use Python for model training, feature importance, and open-deal scoring.

# 4. How To Explain The Analysis

## Core story

- ? Marketing has two contribution views: sourced pipeline and influenced pipeline.
- ? Sourced pipeline is conservative because it only counts deals where CRM says marketing created the deal.
- ? Influenced pipeline is broader because it counts accounts that had marketing touchpoints before opportunity creation.
- ? Email tends to start journeys. 6sense display often appears later and supports sustained visibility.
- ? Coverage is the biggest growth lever: many target accounts are still unreached.
- ? Pipeline volume is growing, but recent cohort win rate is declining, so quality must be protected.

## Recommended actions

- ? Reach strong-fit unreached accounts first.
- ? Start with email outreach, then test 6sense overlay.
- ? Report sourced and influenced pipeline side by side.
- ? Use win probability bands in sales review.
- ? Scale creative patterns that improve engagement, but test causality before large budget shifts.

# 5. Quick Excel Workbook Structure

## Suggested tabs

- ? 00\_ReadMe - project purpose and definitions.
- ? 01\_Raw\_Checks - quick inspection of raw files, not source of truth.
- ? 02\_Output\_Imports - imported generated Excel outputs from outputs/analysis.
- ? 03\_Pivots - PivotTables for channel, segment, creative, email, coverage, and cohort views.
- ? 04\_Dashboard - optional Excel dashboard for manual exploration.
- ? 05\_Notes - business interpretation and presentation talking points.

## Validation before presenting

- ? Run 06\_validate\_metrics.py and confirm it passes with 0 errors.
- ? Confirm public/index.html matches the generated dashboard.
- ? Use Marketing\_Analytics\_Executive\_Deck\_v4.pptx as the final deck.
- ? Check that AUC, coverage, pipeline, and attribution numbers match the regenerated outputs.